MATEO ZITELLA MINI

(203) 559-4075 · zitella.m@northeastern.edu · https://mateozitella.github.io/Portfolio/ · https://www.linkedin.com/in/mateozitella/

EXPERIENCE

Schleifring North America | Lowell, MA August 2025 - Present
Manufacturing Engineer Coop
- Create Solid Edge models, technical drawings, and SLA3D print new and existing assembly fixtures
- Evaluate multiple faulty slip rings daily using a microscope and multimeter to assign appropriate repair and quality test tasks to people in assembly and quality control through SAP
- Collaborate with the machine shop to create clear technical drawings and easily manufacturable designs
- Write and improve existing work instructions for assembly and rework of slip rings
- Update the format and contents of a company-wide fixture CAD tracking database in Excel

Northeastern DeLTA Lab | Burlington, MA October 2024 - July 2025
Research Assistant
- Designed equipment for experiments involving X-ray optical testing of different material properties
- Referenced technical drawings to design adjustable fixtures for lab equipment that interface with the optical table
- Designed a safety cover for laser experiments in SolidWorks using 80/20 bars and acetal sheets

Northeastern Silicon Synapse Lab | Boston, MA October 2024 - May 2025
Research Assistant for Aerobat Project
- Designed and prototyped a print-in-place design to increase ease of assembly for Aerobat, a flapping wing robot
- Planned toolpaths in Fusion360 to CNC machine lightweight aluminum gears for Aerobat
- Assembled and soldered electrical components to minimize the weight of wires

PROJECTS

Combat Robotics Club
Stormbringer - 3 Pound Combat Robot January 2025 - June 2025
- Researched successful combat robot designs and compared different ideas for the robot's weapon system
- Investigated compatibility of electronics including a weapon motor, weapon ESC, drive ESC, and LiPo battery
- Designed the body and weapon, taking mechanical hardware, materials, and weight optimization into account

GALINT - 1 Pound Combat Robot September 2024 - March 2025
- Designed iterations of the robot in SolidWorks to optimize performance while meeting the weight requirement
- 3D printed, soldered, and assembled the robot's frame and electronics
- Used hand drills and a dremel to make changes to the robot's frame and weapon after being 3D printed

Forge Product Development Lab
Smart Step September - December 2024
- Developed a telescopic actuator mechanism using Solidworks for a height-changing smart cane, designed the body tube linkage system, and troubleshot electronics and C++ code
- Collaborated with a team of 7 to select lightweight and cost efficient hardware
- Created an Onshape assembly and edited team member's designs to ensure integration into the overall design

NASA Student Launch Challenge
Subscale Rocket September - November 2024
- Collaborated with a team of 7 people to design fin mounts for a 4.85′ subscale rocket in SolidWorks and Onshape
- Designed and simulated a subscale rocket model using OpenRocket to analyze flight performance and stability

EDUCATION

Northeastern University College of Engineering | Boston, MA May 2027
Candidate for Bachelor of Science in Mechanical Engineering **GPA: 3.57** (Dean's List)
- *Relevant Courses*: Fluid Mechanics, Dynamics, Intro to Material Science, Mechanics of Materials, Thermodynamics, Statics, Differential Equations & Linear Algebra, Physics 1-2, Calculus 1-3

SKILLS

Software: SolidWorks, Solid Edge, Onshape, Fusion360 CAM, SAP, ECTR, Adobe Illustrator, OpenRocket, Microsoft Office, Google Suite
Fabrication: Arduino, Soldering, SLA and FDM 3D printing, Laser cutting, CNC mill, Woodworking, Hand tools
Languages: Spanish, C++, Python, CSS, HTML, JavaScript, MATLAB
Interests: Art, Guitar, Anime, Product Development